TOMÁS ELÍAS VALDERRAMA PLATZ

Data Scientist | Oceanography & Environmental Science | M.Sc. in Geophysics
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PROFESSIONAL PROFILE

Data scientist with a background in geophysics and experience in processing environmental, oceanographic, and satellite data. Proficient in Python, statistical analysis, and data visualization, with a proven ability to automate workflows and build data pipelines. Focused on turning complex data into actionable insights for technical, academic, and regulatory organizations.

TECHNICAL SKILLS

Advanced: Python (pandas, numpy, matplotlib, cartopy, geopandas, shapely, xarray, netCDF4), MATLAB, Bash, Git/GitHub, Excel/Google Sheets, ArcGIS, QGIS
Intermediate: Google Earth Engine (JavaScript), React/JavaScript (HTML/CSS), R
Numerical Models: CROCO-ROMS, OpenDrift, ILLUMINA, WRF, SWAN (learning)

PROFESSIONAL EXPERIENCE

Project Engineer - Pelícanos SpA *Dec 2025 – Present*
Environmental engineering and applied oceanography · Data analysis.

- Processing and analysis of wind (including ERA5 reanalysis integration for annual studies), ADCP marine current (Nortek Signature), dissolved oxygen, and salinity data; technical reporting in compliance with NCh 1821-2020.
- Development of Python code to automate analysis workflows and generate maps with ESRI satellite imagery, replacing a manual process in Global Mapper and PowerPoint.
- Delivery of technical reports to aquaculture clients: wind analysis (1-month and annual), ADCP current analysis (1-month and 6-month), and dissolved oxygen and salinity monitoring equipment.
- Participation in field campaigns to retrieve data from wind and water quality instruments.

Data Scientist (Consultant) - LATLON SpA *Jul 2025 – Present*
Project-based consulting in data science and remote sensing.

- *Satellite pipeline (Jul–Aug 2025; client: FONDART researchers, U. de Chile):* Design and implementation of a bulk Sentinel-2 and MODIS image download workflow via Google Earth Engine (JavaScript/GEE); 3-year coverage for the Valparaíso, Metropolitan, and O'Higgins regions; delivery of documented code for project continuity.
- *Light pollution model (Oct–Dec 2025; clients: AURA and OPCC):* Installation and configuration of the ILLUMINA model; preparation of an installation manual; design and execution of *benchmark* experiments using Hawaii data; definition of input/output structure and reference figure for a night-sky monitoring web platform in northern Chile.
- *Specialized Analyst and Scientific Advisor — Exploradores 2026 Project, Ministry of Energy (Apr 2026 – present):* Scientific validation of sup3rwind (NREL), a GAN-based model downscaling ERA5 to 2 km / 5 min for South America (2005–2024); three-level validation framework (sup3r vs. in-situ, WRF vs. in-situ, sup3r vs. WRF); programmatic data access via NREL HSDS API (h5pyd). Metrics: bias, RMSE, Weibull (k, c), diurnal/seasonal cycles, P90/P99 percentiles. Advisory on storage and integration of large-scale wind datasets in Chile.
- *Meteorological study for transmission line — Linieros, northern Chile (Apr 2026 – present):* Extreme wind and temperature analysis (Gumbel distribution, 50-year return period, IEC 60826); ice sleeve thickness estimation using the Goodwin v4 model; data sources: API Exploradores de Energía and ERA5-Land reanalysis.

Graduate Research Assistant *2021 – 2025*
Universidad de Concepción — FONDECYT Regular N°1211230: "Modelling the role of biogeochemical river input into fjord systems in Chilean Patagonia"

- Analysis of the semi-permanent eddy dynamics in the Gulf of Ancud using CROCO-ROMS ocean model outputs: vertical structure, temporal variability, and tidal role.
- Lagrangian particle transport simulations with OpenDrift at multiple depths (0–250 m), with and without tidal forcing.
- Degree awarded (2025). Thesis-derived manuscript in preparation for submission to an indexed international journal.

Internship - IFOP (Instituto de Fomento Pesquero) *2020*
Castro, Chiloé, Chile

- Comparative analysis of streamflow from the VIC hydrological model against 33 freshwater sources in the MOSA-ROMS oceanographic model, covering fjords and channels in southern Chile (Los Lagos and Aysén regions).
- Identification of rivers to update and new sources to incorporate into MOSA-ROMS, resulting in a net increase of ~2,800 m^3/s of freshwater input to the system; work integrated into the project's final report (IFOP, 2020).
- Processing and quality control of oceanographic and environmental data in MATLAB.

EDUCATION

M.Sc. in Geophysics (Modeling and Data Analysis) *2021 – 2025*
Universidad de Concepción, Chile

B.Sc. in Geophysics *2015 – 2021*
Universidad de Concepción, Chile

SELECTED PROJECTS

- **Monitoring Map Generator** — Python tool (~1,500 lines) that integrates KMZ, shapefiles, and JSON to generate multi-page cartographic maps (JPG/PDF) with equipment auto-discovery and ERA5 overlay. Replaced a manual workflow using three tools (Global Mapper + PowerPoint).
- **AFP Fund Evaluator** (live demo) — React/Recharts web app comparing historical performance of pension funds A–E with savings projections, Monte Carlo simulation, and Sharpe ratio. Data from 2002–2024 validated against Superintendencia de Pensiones records.
- **GEE Satellite Pipeline** — Reproducible bulk download workflow for Sentinel-2 and MODIS images via Google Earth Engine; multi-year spatial coverage analysis for three Chilean regions with MGRS tiling in Python.
- **ILLUMINA** — Installation, configuration, and benchmarking of the radiative transfer model for light pollution monitoring; northern Chile.
- **Freshwater inputs to MOSA-ROMS** — Comparative analysis of VIC streamflow vs. oceanographic model sources for fjords in southern Chile; results integrated into IFOP technical report P-656118 (2020).

LANGUAGES

- English: Advanced — technical reading, writing, and oral comprehension
- Spanish: Native